Data Flow Diagram Review

Assessment of the supplied DFD against the current local codebase • 21 October 2026

1. Overall verdict

Needs several corrections.

The diagram identifies the principal users and many real functions. Its data stores and several flows reflect an older model. Active, completed, and cancelled deliveries are rows in one public.assignments table, not separate Dispatches and Deliveries tables.

The diagram is a hybrid DFD/system flowchart, closest to a Level 0 DFD with several top-level processes. Numbering 1.0–8.0 can be acceptable under a course convention; 1–8 is another common choice. A context diagram would instead have one whole-system process and no internal stores.

Scope: repository implementation and migrations were inspected. The live connected Supabase schema was not queried, so migration deployment is not independently confirmed.

2. What is already correct

- OPM, CSR, and Driver are the three principal external users. Use “CSR” rather than “Administrator (CSR).” A legacy admin role still exists with CSR-like operational permissions.
- Authentication, fleet records, dispatching, delivery tracking, formal documents, reports, and audit history are genuine system functions.
- The Driver receives assignments, confirms delivery stages, supplies operational photos, uploads formal documents, and views Past Deliveries.
- Formal documents and the activity log correspond to real persisted records. Reports and dashboards are calculated views and need no separate report store.

Current process

Assessment - Recommended wording or scope

1.0 Login

Partially correct - Authenticate Users; OPM, CSR, and Driver use Supabase Auth.

2.0 CSR Accounts

Partially correct - Manage CSR Accounts and Reset Requests.

3.0 Drivers & Trucks

Mostly correct - Manage Fleet; OPM manages, CSR views availability.

4.0 Dispatch Trucks

Mostly correct - Create and Manage Assignments; include eligible edit/cancel.

5.0 Track Deliveries

Partially correct - Manage Delivery Lifecycle; include stages, evidence, completion.

6.0 Manage Documents

Partially correct - Upload and Review Delivery Documents; Driver uploads, CSR reviews.

7.0 Monitor & Report

Mostly correct - Generate Dashboard and Reports from operational records.

8.0 Activity Log

Partially correct - Record and View Audit Events, or show direct audit writes to its store.

3. What needs to change

Current: “Administrator (CSR)” is a single actor.
Recommended: Label it “CSR”; retain OPM and Driver.
Why: CSR is the current operational role. OPM has separate fleet/account capabilities.

Current: D3 Dispatches and D4 Deliveries imply separate stores.
Recommended: One Assignments store for dispatch details, physical stages, document lifecycle linkage, timestamps, and active/completed/cancelled state.
Why: The UI splits records from public.assignments by its state column; Past Deliveries reads completed rows from it.

Current: Login connects mainly to CSR and D1.
Recommended: Draw credentials from all three actors into Authenticate Users, with result/session and permitted role data flowing back.
Why: All three authenticate through Supabase Auth; staff roles are in profiles and Driver identity links to drivers.

Current: OPM supplies fleet details, but CSR's fleet information is unclear.
Recommended: OPM supplies management data; Manage Fleet returns driver/truck availability to CSR.
Why: CSR can view operational fleet data but cannot register/archive or manage PINs.

Current: “Review & approval” from CSR is not separated from OPM's document access.
Recommended: CSR supplies decisions and rejection reasons; both roles may receive document information, but OPM is currently read-only on operational/document pages.
Why: The current admin shell applies read-only mode to OPM there.

Current: “POD upload” represents all evidence, while photos are absent.
Recommended: Separate formal-document and operational-photo flows.
Why: They use distinct records: assignment_documents and status_photos, with files in Supabase Storage.

Current: Activity Log sends results to OPM but has no clear event inputs.
Recommended: Connect auditable account, fleet, assignment, and document events to Record and View Audit Events, then to/from the Activity Log store.
Why: The log must have writers as well as readers; some events are written by database functions.

Current: Several arrow labels are actions such as “Login” or “Review & approval.”
Recommended: Label arrows with data: credentials, role result, stage confirmation/timestamp, document decision/reason, report criteria/results.
Why: DFD arrows carry data; process names describe transformations.

4. Missing parts

- Operational photos and capture time/location metadata.
- Persisted notifications for milestones, cancellations, and document review, including read state. A store is appropriate; a separate process is optional.
- CSR password-reset requests, OPM decisions, expiry, and completion. Do not show passwords or reset tokens flowing to OPM.
- Delay and issue reports as inputs to the delivery lifecycle.
- Final completion and Past Deliveries. Physical Delivered is followed by document review; final Completed is stored on the assignment. “Ready to Complete” is an eligibility condition, not a separate persisted status.
- Eligible cancellation, reason, and audit entry.

A client/customer name is stored, but that alone does not make the client an external actor in this app's DFD.

5. Corrected DFD structure

Use this as a redraw guide for a Level 0 logical DFD. D4 and D5 each combine database metadata with the associated stored file at this abstraction level.

External entities

Operational Manager (OPM) • CSR • Driver

Processes

1.0 Authenticate Users
2.0 Manage CSR Accounts and Password Reset Requests
3.0 Manage Fleet
4.0 Create and Manage Assignments
5.0 Manage Delivery Lifecycle and Evidence
6.0 Upload and Review Delivery Documents
7.0 Generate Dashboard and Reports
8.0 Record and View Audit Events

Data stores

D1 Auth Identities and Staff Profiles - Supabase Auth, profiles, Driver login linkage/PIN records

D2 Fleet Records - drivers and trucks

D3 Assignments - active/completed/cancelled, route, stage, timestamps

D4 Operational Photo Evidence - status_photos and Storage files

D5 Formal Delivery Documents - assignment_documents and Storage files

D6 Notifications - notifications

D7 Activity Log - activity_log

D8 CSR Password Reset Requests - csr_password_reset_requests

Data flows: authentication and accounts

OPM / CSR / Driver → 1.0: login credentials
1.0 ↔ D1: identity and role verification data
1.0 → OPM / CSR / Driver: authentication result and permitted session
OPM → 2.0: CSR account details; reset approval or rejection
CSR → 2.0: reset request; new password through approved reset flow
2.0 ↔ D1: CSR identity/profile and password change
2.0 ↔ D8: reset request, decision, expiry, completion
2.0 → OPM: member list and request status
2.0 → CSR: generic request confirmation and reset outcome

Data flows: fleet and assignments

OPM → 3.0: driver/truck registration, archive, PIN changes
3.0 ↔ D2: driver, truck, availability, and PIN data

3.0 → OPM: fleet management result
3.0 → CSR: driver/truck operational availability
CSR → 4.0: dispatch details; eligible edit/cancellation details
D2 → 4.0: available driver and truck data
4.0 ↔ D3: assignment details and cancellation state/reason
4.0 → CSR: saved assignment and action result

Data flows: delivery and evidence

D3 → 5.0: assigned task, stage, and delivery history
5.0 → Driver: current task, addresses, status, Past Deliveries
Driver → 5.0: stage confirmations, photo evidence, delay/issue details
5.0 ↔ D3: stage/status changes, timestamps, completion state
5.0 ↔ D4: operational photo records and files
5.0 → CSR / OPM: status and evidence for monitoring
CSR → 5.0: permitted final completion action
5.0 → D6: milestone/cancellation notice data

Data flows: documents, reports, and audit

Driver → 6.0: required formal files and corrected uploads
6.0 ↔ D5: files, upload times, review status
D3 → 6.0: assignment and physical-delivery eligibility
CSR → 6.0: approval/rejection decision and reason
6.0 → Driver: review/correction status
6.0 → CSR / OPM: files and review status
6.0 → D6: documents-ready or correction notice data
D2 / D3 / D5 / D6 → 7.0: fleet, assignment, document, notice data
CSR / OPM → 7.0: dashboard or report selection
7.0 → CSR / OPM: dashboard figures and reports
2.0 / 3.0 / 4.0 / 5.0 / 6.0 → 8.0: auditable event data
8.0 ↔ D7: recorded events and requested history
OPM → 8.0: activity-log search/filter criteria
8.0 → OPM: activity-log results

For notification presentation, connect D6 through an existing user-facing process to CSR/OPM. If that clutters the drawing, a ninth process named “Present Notifications” is reasonable. Never draw an entity-to-store or store-to-store arrow directly.

6. Final recommendation

Redraw several parts. Keep the three-actor layout and most process positions. Replace the store model, separate photos from formal documents, add notifications and reset requests, and relabel arrows as data. The supplied image should not be submitted as an accurate representation of the current implementation without these changes.

Repository sources inspected: src/app/lib/db.ts; src/app/App.tsx; src/app/lib/staffPermissions.ts; src/app/lib/assignmentWorkflow.ts; src/app/lib/notificationEvents.ts; src/app/components/driver/DriverInterface.tsx; src/app/components/admin/AdminDashboard.tsx; src/app/components/admin/Reports.tsx; Supabase migrations for the relational schema, OPM role, cancellation, workflow, and CSR password reset.